Note on Fermionizing Virasoro Minimal Models

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Abstract

This note gives some advanced idea in fermionizing the Virasoro Minimal Models.

Contents

1 Ising CFT and Majorana CFT Dual

1.1 Background Gauge Field Language

The bosonization of Ising and Majorana CFT are commonly written in a background gauge field language. Here we give a brief, nonrigorous introduction.

1.1.1 $\mathbb{Z}_2$ Gauge Field

1.1.1.1 $\mathbb{Z}_2$ Gauge Theory

Boundary conditions in the bosonic theory can be understood in a unified way using the language of a $\mathbb{Z}_2$ gauge theory. We introduce a background gauge field

$$A \in H^1(\Sigma_g, \mathbb{Z}_2). \tag{1}$$

This object admits several equivalent interpretations:
- as a cohomology class,
- as a flat $\mathbb{Z}_2$ connection,
- as a homomorphism from the fundamental group, $A \in \mathrm{Hom}(\pi_1(\Sigma_g), \mathbb{Z}_2)$.

Concretely, A assigns a $\mathbb{Z}_2$ value to each non-contractible cycle. Choosing a canonical homology basis $\{a_1, b_1, ..., a_g, b_g\}$, we write

$$A(a_i) = \alpha_i \in \mathbb{Z}_2, \quad A(b_i) = \beta_i \in \mathbb{Z}_2, \tag{2}$$

so the gauge field is completely specified by

$$A = (\alpha_1, \beta_1, ..., \alpha_g, \beta_g) \in (\mathbb{Z}_2)^{2g}. \tag{3}$$

1.1.1.2 Cup Product

Given two $\mathbb{Z}_2$ gauge fields $A, B \in H^1(\Sigma_g, \mathbb{Z}_2)$, their cup product is an element

$$A \cup B \in H^2(\Sigma_g, \mathbb{Z}_2) \simeq \mathbb{Z}_2. \tag{4}$$

On a closed Riemann surface this evaluates to a number:

$$\int_{\Sigma_g} A \cup B \in \mathbb{Z}_2. \tag{5}$$

In terms of the cycle basis, writing $A = (\alpha_i, \beta_i)$ and $B = (\alpha'_i, \beta'_i)$,

$$\int_{\Sigma_g} A \cup B = \sum_{i=1}^{g} (\alpha_i \beta'_i - \beta_i \alpha'_i) \mod 2. \tag{6}$$

Since $-1 \equiv 1$ in $\mathbb{Z}_2$, this simplifies to

$$\int_{\Sigma_g} A \cup B = \sum_{i=1}^{g} (\alpha_i \beta'_i + \beta_i \alpha'_i) \mod 2. \tag{7}$$

For the torus T^2 the cup product table is:

$\int_{T^2} A \cup B$	$A_0 = (0,0)$	$A_1 = (1,0)$	$A_2 = (0,1)$	$A_3 = (1,1)$
$A_0 = (0,0)$	0	0	0	0
$A_1 = (1,0)$	0	0	1	1
$A_2 = (0,1)$	0	1	0	1
$A_3 = (1,1)$	0	1	1	0

Table 1: Cup product integral on T^2 for the four $\mathbb{Z}_2$ gauge fields, computed modulo 2.

1.1.1.3 Bosonic Partition Function Coupled to Background Field

Partition functions with different boundary conditions of a bosonic theory (or a $\mathbb{Z}_2$ defect-twisted theory) can be understood as coupling to different background $\mathbb{Z}_2$ gauge fields:

$$Z[A], \quad A \in H^1(\Sigma_g, \mathbb{Z}_2), \tag{8}$$

where A specifies the twisting of fields along non-contractible cycles. The conventions are:

- 0 for Periodic Boundary Condition (did not cross a defect),
- 1 for Anti-Periodic Boundary Condition (crossed a defect).

For example, on the torus $\Sigma_1 = T^2$ with cycles (a, b) we write $A = (\alpha, \beta) \in \mathbb{Z}_2 \times \mathbb{Z}_2$, and the four boundary conditions are

$$Z_{P,P} = Z[(0,0)], \quad Z_{P,\mathrm{AP}} = Z[(0,1)], \quad Z_{\mathrm{AP},P} = Z[(1,0)], \quad Z_{\mathrm{AP},\mathrm{AP}} = Z[(1,1)]. \tag{9}$$

◊ YL: In fact I don't understand why coupling to a gauge field is equal to inserting a Symmetry defect line; nevertheless, the calculational result is the same in partition functions.

1.1.2 Spin Structure

1.1.2.1 Definition

Mathematically, a spin structure can be defined via the quadratic refinement of a $\mathbb{Z}_2$ gauge field,

$$\rho : H^1(\Sigma_g, \mathbb{Z}_2) \to \mathbb{Z}_2. \tag{10}$$

Physically, we simply understand it as specifying the boundary conditions of fermions along the non-contractible cycles. As with the $\mathbb{Z}_2$ gauge field, we assign a label to each cycle:

- 1 for Periodic Boundary Condition (Ramond, R),
- 0 for Anti-Periodic Boundary Condition (Neveu-Schwarz, NS).

> **Remark**
>
> This convention is **opposite** to the bosonic case: there, 0 denotes periodic.

On the torus T^2, a spin structure determines whether the fermion is periodic or antiperiodic along each cycle:

$$\rho = (\text{a-cycle}, \text{b-cycle}) \in \{\mathrm{AP}, \mathrm{P}\}^2. \tag{11}$$

Two spin structures are related by a background $\mathbb{Z}_2$ gauge field $A \in H^1(\Sigma_g, \mathbb{Z}_2)$ via

$$\rho + A = (\rho_a + A_a \mod 2, \quad \rho_b + A_b \mod 2), \tag{12}$$

which physically corresponds to flipping the fermion boundary condition along cycles where A is nonzero.

1.1.2.2 Arf Invariant

The Arf invariant is a $\mathbb{Z}_2$-valued number depending only on the spin structure; it distinguishes "even" from "odd" spin structures. Mathematically it is defined using quadratic refinements acting on basis elements.

On T^2 with standard cycles (a, b) we label spin structures with 0 (AP) and 1 (P). The four spin structures and their Arf invariants are:

Spin structure (ρ_a, ρ_b)	**a-cycle**	**b-cycle**	**Arf(ρ)**
$(0, 0)$	0	0	0
$(1, 0)$	1	0	0
$(0, 1)$	0	1	0
$(1, 1)$	1	1	1

Table 2: Arf invariants of the four spin structures on T^2.

1.1.3 Useful Results Involving the Arf Invariant

Cup product calculation, For all $\mathbb{Z}_2$ gauge fields $s \in H^1(\Sigma_g, \mathbb{Z}_2)$:

$$\frac{1}{2^g} \sum_s (-1)^{\int s \cup t} = \begin{cases} 2^g & \text{if } t = 0 \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

Arf Invariant of spin structure and background gauge field interacting:

$$\mathrm{Arf}[a + b + \rho] = \mathrm{Arf}[a + \rho] + \mathrm{Arf}[b + \rho] + \mathrm{Arf}[\rho] + \int a \cup b. \tag{14}$$

Arf invariant and spin structure sum:

$$\frac{1}{2^g} \sum_S (-1)^{\mathrm{Arf}(s+\rho) + \ \mathrm{Arf}[\rho] + \int s \cup t} = (-1)^{\mathrm{Arf}[t+\rho]}. \tag{15}$$

1.2 Ising/Majorana CFT Fermionization Duality

The Ising CFT and Majorana CFT are related by a web of dualities. We use the torus partition function as an illustrative example; the dualities are claimed to hold on general closed Riemann surfaces.

Here is a diagram illustrating the dualities:

$\mathcal{F}$	$(-1)^F$-even	$(-1)^F$-odd
NSNS	$1, i\psi_\mathrm{L}\psi_\mathrm{R}$	$\psi_\mathrm{L}, \psi_\mathrm{R}$
RR	σ	μ

$\xleftrightarrow{\otimes(-1)^{\mathrm{Arf}}}$

$\mathcal{F} \otimes (-1)^{\mathrm{Arf}}$	$(-1)^F$-even	$(-1)^F$-odd
NSNS	$1, i\psi_\mathrm{L}\psi_\mathrm{R}$	$\psi_\mathrm{L}, \psi_\mathrm{R}$
RR	μ	σ

fermionization $\updownarrow$ bosonization

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$\mathcal{B}$	$\mathbb{Z}_2$-even	$\mathbb{Z}_2$-odd
untwisted	$1, \varepsilon$	σ
twisted	μ	$\psi_\mathrm{L}, \psi_\mathrm{R}$

$\xleftrightarrow{\mathbb{Z}_2 \text{ orbifold}}$

$\mathcal{B}/\mathbb{Z}_2$	$\mathbb{Z}_2$-even	$\mathbb{Z}_2$-odd
untwisted	$1, \varepsilon$	μ
twisted	σ	$\psi_\mathrm{L}, \psi_\mathrm{R}$

Figure 1: Duality web relating Ising CFT and Majorana CFT.

1.2.1 Partition Function Data

We collect the partition functions with explicit spin structure (fermionic theory) or background gauge field (bosonic theory) together with their character expansions.

Partition Function	**Spin Structure**	**Characters**
$Z_{\mathrm{AP,AP}}$	$Z[(0,0)]$	$\lvert\chi_0\rvert^2+\lvert\chi_{1/2}\rvert^2+\overline{\chi}_0\chi_{1/2}+\overline{\chi}_{1/2}\chi_0$
$Z_{\mathrm{AP},P}$	$Z[(0,1)]$	$\lvert\chi_0\rvert^2+\lvert\chi_{1/2}\rvert^2-\overline{\chi}_0\chi_{1/2}-\overline{\chi}_{1/2}\chi_0$
$Z_{P,\mathrm{AP}}$	$Z[(1,0)]$	$2\lvert\chi_{1/16}\rvert^2$
$Z_{P,P}$	$Z[(1,1)]$	0

Table 3: Partition functions and characters for Majorana CFT.

Partition Function	**BG Gauge Field**	**Characters**
$Z_{P,P}$	$Z[(0,0)]$	$\lvert\chi_0\rvert^2+\lvert\chi_{1/2}\rvert^2+\lvert\chi_{1/16}\rvert^2$
$Z_{P,\mathrm{AP}}$	$Z[(0,1)]$	$\lvert\chi_0\rvert^2+\lvert\chi_{1/2}\rvert^2-\lvert\chi_{1/16}\rvert^2$
$Z_{\mathrm{AP},P}$	$Z[(1,0)]$	$\overline{\chi}_0\chi_{1/2}+\overline{\chi}_{1/2}\chi_0+\lvert\chi_{1/16}\rvert^2$
$Z_{\mathrm{AP,AP}}$	$Z[(1,1)]$	$-\overline{\chi}_0\chi_{1/2}-\overline{\chi}_{1/2}\chi_0+\lvert\chi_{1/16}\rvert^2$

Table 4: Partition functions and characters for Ising CFT.

1.2.2 Ising = Ising/$\mathbb{Z}_2$ (Kramers-Wannier Duality)

This duality is a $\mathbb{Z}_2$ orbifold exchanging the A-series and D-series in the minimal model classification. For the Ising CFT it corresponds to the Kramers-Wannier duality, which relates the partition function of the low-temperature phase to that of the high-temperature phase. Since the A-series and D-series are isomorphic for the Ising model, the orbifold gives a transformation between two isomorphic theories.

The traditional perspective proceeds in two steps:

1. **Project** the theory onto the $\mathbb{Z}_2$-even Hilbert space:

$$Z \quad\to\quad \frac{1}{2}\big(Z_{P,P}+Z_{P,\mathrm{AP}}\big). \tag{16}$$

2. **Restore** modular invariance by adding the $\mathbb{Z}_2$-twisted sector Hilbert space:

$$+\frac{1}{2}\big(Z_{\mathrm{AP},P}+Z_{\mathrm{AP,AP}}\big). \tag{17}$$

This gives the same partition function as the Ising CFT with PP boundary conditions:

$$Z_{\mathrm{orb}}=\frac{1}{2}\big(Z_{P,P}+Z_{P,\mathrm{AP}}+Z_{\mathrm{AP},P}+Z_{\mathrm{AP,AP}}\big)=Z_{P,P}. \tag{18}$$

Now we use the background gauge field language to give a more general discussion of this gauging process.

Theorem 1.1 **$\text{Ising} = \text{Ising}/\mathbb{Z}_2$**

Coupling to a background gauge field A, the partition functions of the two theories are related by

$$Z_{\text{Ising}/\mathbb{Z}_2}[A] = \sum_{a \in H^1(\Sigma_g, \mathbb{Z}_2)} Z_{\text{Ising}}[a](-1)^{\int a \cup A}. \tag{19}$$

Example. For $A = (0,0)$:

$$Z_{\text{Ising}/\mathbb{Z}_2}[(0,0)] = \frac{1}{2} \sum_{a \in H^1(\Sigma_g, \mathbb{Z}_2)} Z_{\text{Ising}}[a] = Z_{\text{Ising}}[(0,0)]. \tag{20}$$

For $A = (0,1)$:

$$\begin{aligned} Z_{\text{Ising}/\mathbb{Z}_2}[(0,1)] &= \frac{1}{2}\big(Z_{\text{Ising}}[(0,0)] + Z_{\text{Ising}}[(0,1)] - Z_{\text{Ising}}[(1,0)] - Z_{\text{Ising}}[(1,1)]\big) \\ &= Z_{\text{Ising}}[(0,1)]. \end{aligned} \tag{21}$$

One can verify from these calculations that the two theories are isomorphic.

◊ YL: In fact, this story have more to tell, the dual model have a SM realization and it can be seen in SM theory, moreover, we can also see this and inverse process in CFTs, by gauging a dual symmety. But I'm not familiar with these thus not talking about them here.

1.2.3 Majorana $=$ (Ising $\otimes$ Arf$/\mathbb{Z}_2$ (Fermionization)

Theorem 1.2 **Majorana $=$Ising $\otimes$ Arf$/\mathbb{Z}_2$**

$$Z_{\text{Majorana}}[A+\rho] = \frac{1}{2^g}(-1)^{\text{Arf}[\rho]} \sum_{a \in H^1(\Sigma_g, \mathbb{Z}_2)} Z_{\text{Ising}}[a](-1)^{\text{Arf}[a+\rho] + \int a \cup A}. \tag{22}$$

Example. Take $A = 0$ and fix the spin structure $\rho = (0,0)$ (two NS boundary conditions). Since $\text{Arf}[(0,0)] = 0$, the sign $(-1)^{(\text{Arf}[a+\rho]\}}$ equals -1 only for $a = (1,1)$. Therefore:

$$\begin{aligned} Z_M[(0,0)] &= \frac{1}{2}(Z[(0,0)] + Z[(1,0)] + Z[(0,1)] - Z[(1,1)]) \\ &= |\chi_0|^2 + |\chi_{1/2}|^2 + \overline{\chi}_0 \chi_{1/2} + \overline{\chi}_{1/2}\chi_0. \end{aligned} \tag{23}$$

This matches the expected result from the Majorana partition function table exactly.

1.2.4 Ising $=$ Majorana$/\mathbb{Z}_2$ (Bosonization)

Theorem 1.3 **Ising $=$ Majorana$/\mathbb{Z}_2$**

$$Z_{\text{Ising}}[A] = \frac{1}{2^g}(-1)^{\text{Arf}[A+\rho] + \text{Arf}[\rho]} \sum_{a \in H^1(\Sigma_g, \mathbb{Z}_2)} Z_{\text{Majorana}}[a+\rho](-1)^{\int a \cup A}. \tag{24}$$

Example. Take $A = (0,0)$:

$$\begin{aligned} Z_I[(0,0)] &= \frac{1}{2}(Z_M[(0,0)] + Z_M[(1,0)] + Z_M[(0,1)] + Z_M[(1,1)]) \\ &= |\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2. \end{aligned} \tag{25}$$

1.2.5 Majorana$'$ = Arf $\otimes$ Majorana

One can define a dual theory when considering the Ramond sector of Majorana CFT. When computing the partition function in the Ramond sector, there is a freedom in choosing the fermion parity assignment for the two degenerate ground states $|\sigma\rangle$ and $|\mu\rangle$:

- $(-1)^F\ |\sigma\rangle = +|\sigma\rangle$ and $(-1)^F\ |\mu\rangle = -|\mu\rangle$,
- $(-1)^F\ |\sigma\rangle = -|\sigma\rangle$ and $(-1)^F\ |\mu\rangle = +|\mu\rangle$.

In principle these two choices define two different theories, yet they share the same torus partition function since it does not matter which state is labeled even or odd.

The two theories are related by the Arf invariant partition function (also called the Kitaev chain [1]).

Theorem 1.4 **Majorana$'$ = Arf $\otimes$ Majorana**

$$Z_{\text{Majorana}'}[A+\rho] = (-1)^{\text{Arf}[A+\rho]} Z_{\text{Majorana}}[A+\rho]. \tag{26}$$

This relation is trivially consistent: changing the fermion parity does not change the partition function of Majorana CFT. Indeed, the only nonzero Arf invariant on T^2 occurs for spin structure $(1,1)$, where the partition function is already zero.

One can show that the two theories are related by a chiral transformation. Although chiral symmetry is a classical symmetry, it carries an anomaly that produces exactly:

$$Z[\rho] \quad \to \quad Z[\rho] \times (-1)^{\text{Arf}[\rho]}. \tag{27}$$

See [2] for a detailed explanation.

1.2.6 Majorana$'$ and Ising/$\mathbb{Z}_2$

From the relations above together with mathematical properties of the Arf invariant and the cup product integral, one can prove that Majorana$'$ and Ising/$\mathbb{Z}_2$ satisfy the same duality of partition functions as Ising CFT and Majorana CFT. One can see this using the above dualities and the properties of the Arf invariant and cup product integral.

There is an important result from this duality, is that the Majorana$'$ theory in fact is related to the original Majorana CFT by a chiral transformation. The partition function is different due to the chiral anomaly.

And this means that after gauging the $\mathbb{Z}_2$ symmetry in the Majorana CFT and the Majorana $'$ theory, we get Ising CFT and the orbifolded theory, this means that the Kramers-Wannier duality is in fact a consequence of the chiral anomaly of Majorana CFT.

This is further developed in the language of non-invertible symmetries. The Kramers-Wannier duality is a non-invertible symmetry of the Ising CFT, and it can be seen as a consequence of the chiral anomaly of the Majorana CFT [3].

1.2.7 Final Comments

All the results above are established in [2]. They follow from a "seed duality" which is not proved in that paper but appears to be a long-known result; the proof seems to appear in a mathematical literature.

◊ YL: I have no idea why the Arf invariant is identical to the Kitaev chain. Moreover, I don't quite understand the dual symmetry generated after gauging a symmetry, nor how gauging the dual symmetry returns one to the original theory. These are all very interesting topics but I'm not talking about them here.

2 Fermionization of Virasoro Minimal Models

The above discussion of the Ising CFT and the Majorana CFT can be generalized to general to A series and D series Virasoro minimal models. And they can be fermionized into the corresponding fermionic minimal models. This work is initially given in [4]. Another equilvalent work is given by [5], which is given by a TQFT construction, which efficiently give out the Hilbert Space and the structural constant of Fermionic Minimal Models.

2.1 Fermionization of Minimal Models

This section will review the fermionization of Virasoro minimal model from [4].

2.2 Fermionization from Defects

This section will review the fermionization of Virasoro minimal model from [5].

3 Fermionization of Boundary States

Now with the bulk partition function and Hilbert space give, we turn to boundary cases. For boundary cases, we mainly focus on the boundary states and boundary operators.

3.1 Ising and Majorana CFT Case

The fermionization of Ising CFT and Majorana CFT is well know. The boundary states is given by old papers like [6]. These notes focus on an exact calculation from the Lagrangian formalism of the Majorana CFT.

Here, I'll list out the results of these works and it might be good to compare to modern works:

1. First we take the Lagrangian of Majorana CFT we can explicitly see that the gluing condition gives us two different boundary conditions, free and fix.
2. Then try to do canonical quantization. With the doubling trick, the theory with two different boundary conditions

3.2 Fermionization of Boundary States

A modern work on the fermionization of boundary states is given in [7].

3.3 Anomalies in Boundary Theory

The maually adding of a boundary fermion is an indication of the anomaly of the theory. Another paper on this topic gives a discussion from this perspective [8].

4 Fermionization of Boundary Operators

4.1 Bulk-Boundary OPE from Sewing Constraints

This work is partially given by another way of working with the Fermionic Minimal Models, which is given by [5].

4.2 Solution to Boundary Crossing Equation

This is a work done by DaZhuo Recently.

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